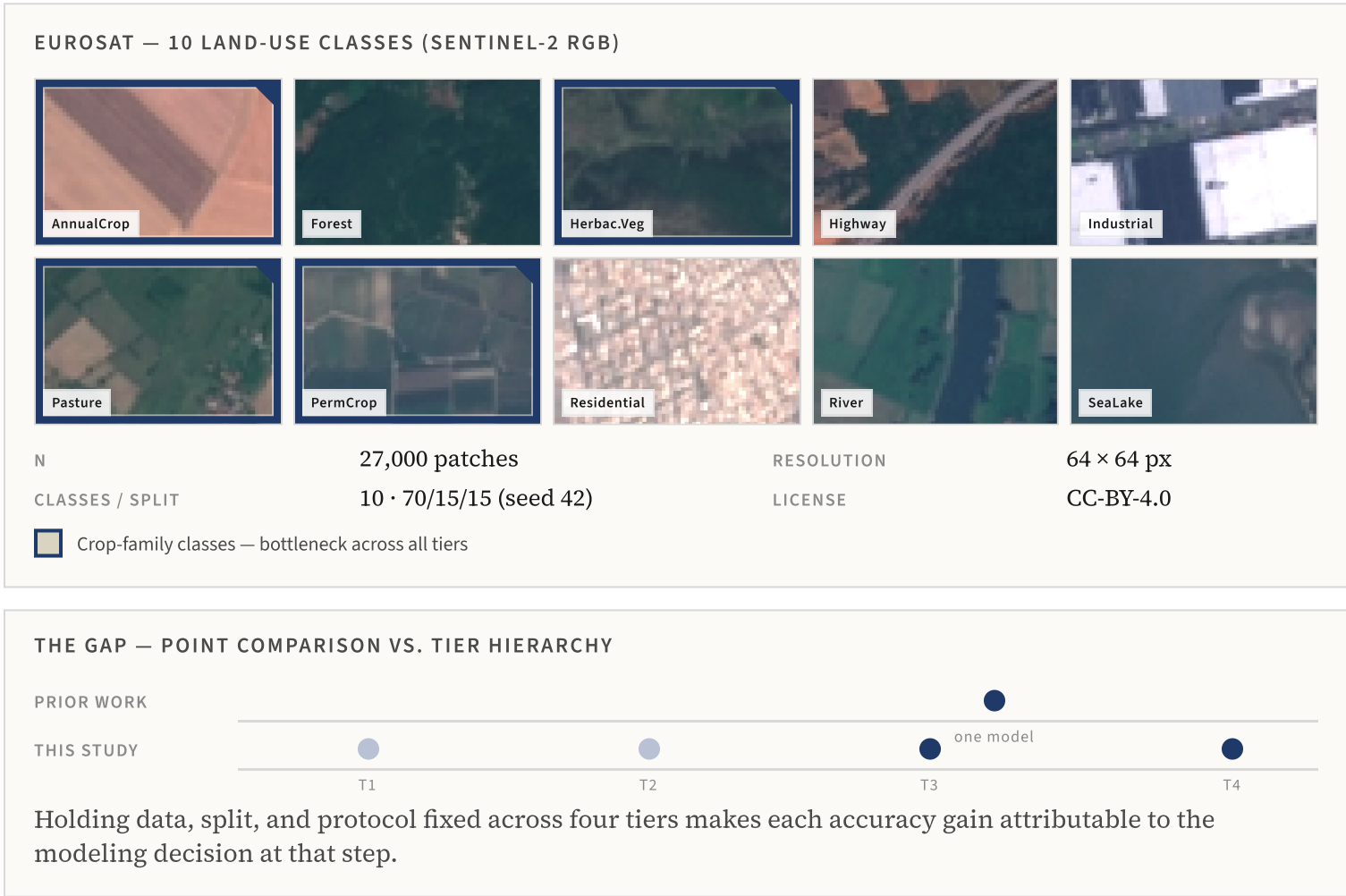


Problem Statement

Land-use classification on EuroSAT RGB.

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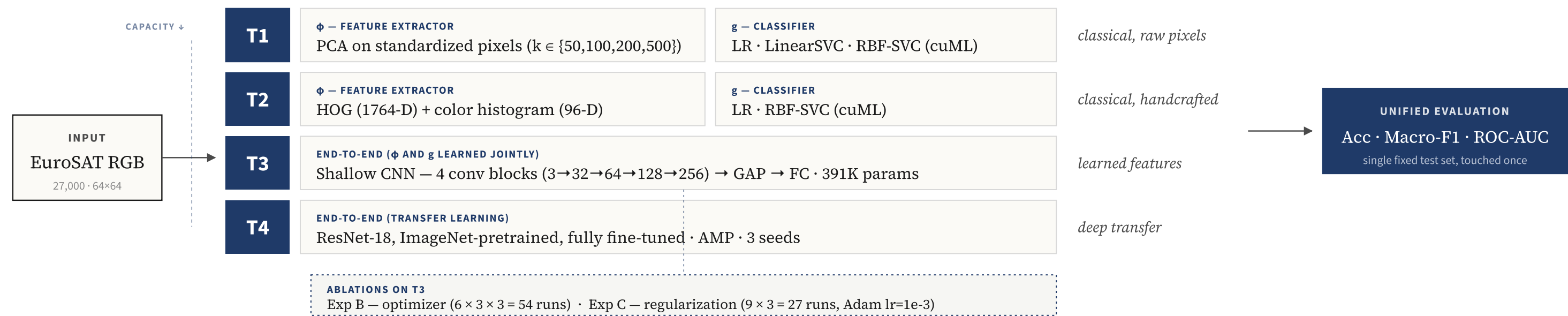
THREE RESEARCH QUESTIONS

- RQ 1** How does test accuracy scale across a tiered hierarchy of feature extractors — raw pixels, handcrafted features, a shallow CNN, and a fine-tuned ResNet-18 — and at which tier do the marginal returns diminish?
- RQ 2** Among six widely-used optimizers, which converges fastest and which generalizes best on the shallow CNN? Does the Wilson et al. (2017) claim that SGD with momentum dominates Adam hold on this dataset?
- RQ 3** Among five regularization techniques (L2, Dropout, BatchNorm, Data Augmentation, Early Stopping), which most reduces the train/test gap on the shallow CNN, and how do they compose when stacked?

Proposed Solution

A controlled four-tier hierarchy on a single fixed split, with two ablations on the shallow CNN.

All four tiers share an identical seed-42 stratified 70 / 15 / 15 split. The test set is touched once per model.



A GPU-resident `TensorDataset` reduces per-epoch time from **105 s** to **1.06 s** ($\approx 100\times$), making the 54-run optimizer ablation tractable on a single A100.

Results & Conclusion

Test-set performance is monotonic across tiers; gains concentrate at the handcrafted-to-learned-features boundary.

TIER COMPARISON (BEST OF EACH TIER, SINGLE FIXED TEST SET)

	BEST MODEL	TEST ACCURACY (60% — 100%)	ACC.	MACRO-F1	Δ
T1	SVM-RBF on PCA(500)		0.6872	0.6826	—
T2	SVM-RBF on HOG + color histogram		0.8309	0.8271	+14.4 pp
T3	Shallow CNN + BatchNorm (3 seeds)		0.9657	0.9648	+13.5 pp
T4	ResNet-18, fine-tuned (3 seeds)		0.9816	0.9812	+1.6 pp

60%

80%

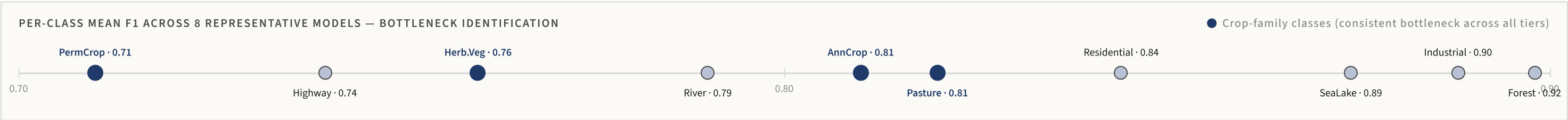
100%

RQ 2 — OPTIMIZER

Top-4 optimizers (Adam, NAG, SGD+Momentum, RMSProp) tied within 0.3 pp. Adam reaches 90% val in 10 epochs vs 13–21 — wins on speed, not asymptote. Wilson et al. (2017) not confirmed.

RQ 3 — REGULARIZATION

BatchNorm alone strongest single regularizer (+1.1 pp). Stacking all five degrades by −3 pp — over-regularization on a clean dataset.



Conclusion. *Representation learning (the T2 → T3 step) is where the accuracy gain lives; once a competent CNN with BatchNorm is in place, additional depth and optimizer choice yield diminishing returns.*

